

Bengali Audio Deepfake Detection

A Research Framework for Robust Forensic Defense

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Presentation Outline

- 1 Problem Definition
- 2 Datasets and Analysis
- 3 Proposed Solutions
- 4 Performance Report
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- 6 Comparative Analysis & Conclusion
- 7 Challenges & Future Directions

The Rising Threat of Audio Deepfakes

- Generative AI (TTS/VC) produces near-indistinguishable synthetic speech
- VITS architecture: End-to-end with stochastic duration predictor + HiFi-GAN decoder + adversarial training
- Removes obvious artifacts (spectral banding, metallic tones)
- Leaves subtle statistical anomalies in phase coherence and high-frequency envelope

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Why Bengali?

- Low-resource language with limited datasets and benchmarks
- Spoken by 250+ million people; heavy use of voice messaging platforms
- Unique phonetics: Dental/retroflex stops, aspirates, complex conjuncts
- Generic detectors fail due to language-specific artifacts

Asymmetric Threat & Research Gap

Technological Asymmetry

Generative models advance rapidly while detection lags — especially for low-resource languages.

Our Focus

Develop a robust Bengali audio deepfake detector that generalizes beyond specific TTS systems (e.g., VITS in BanglaFake).

Key Challenge: Models overfit to artifacts of one generator and fail on unseen TTS (Gemini, Crikk).

Datasets Overview

- **LSTM Bangla Audio Dataset** (baseline): 4,500 samples (2,250 real + 2,250 fake)
- **BanglaFake Dataset**: 26,592 samples (13,796 real + 12,796 VITS fake)
- **TTS Generated Fake Audio Dataset**: 1,921 samples (921 Gemini + 1,000 Crikk)
- **Augmented Evaluation Dataset**: 7,321 samples with diverse real and fake sources

Dataset Distribution Comparison

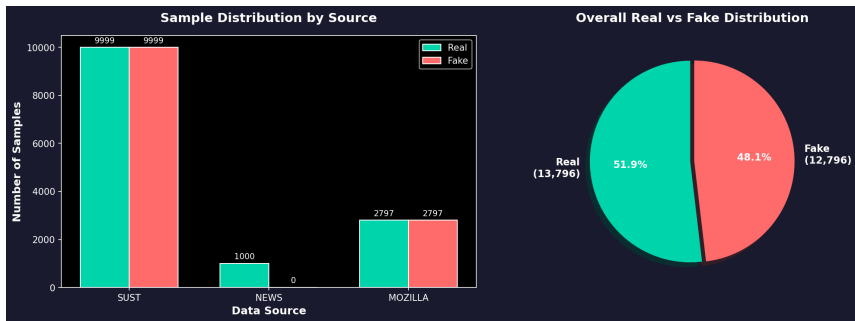


Figure: BanglaFake Dataset

Dataset Distribution Comparison

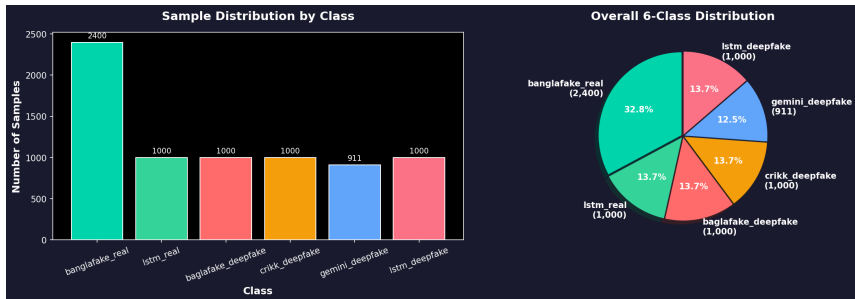


Figure: Augmented Evaluation Dataset

High Frequency Analysis

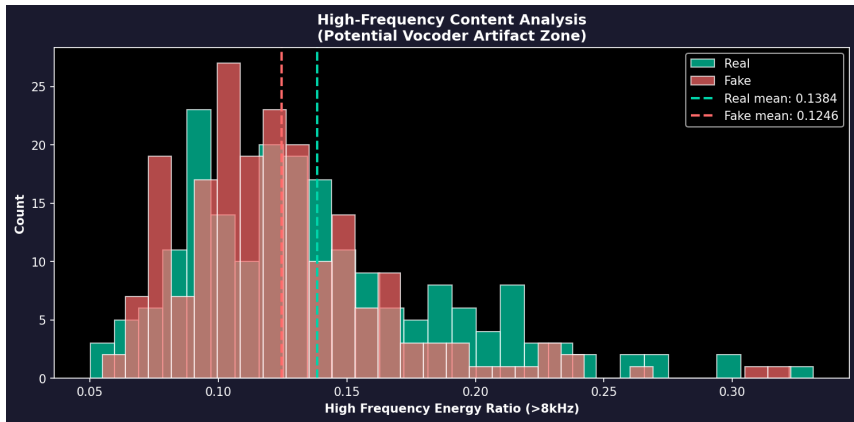


Figure: BanglaFake — High Frequency Analysis

High Frequency Analysis

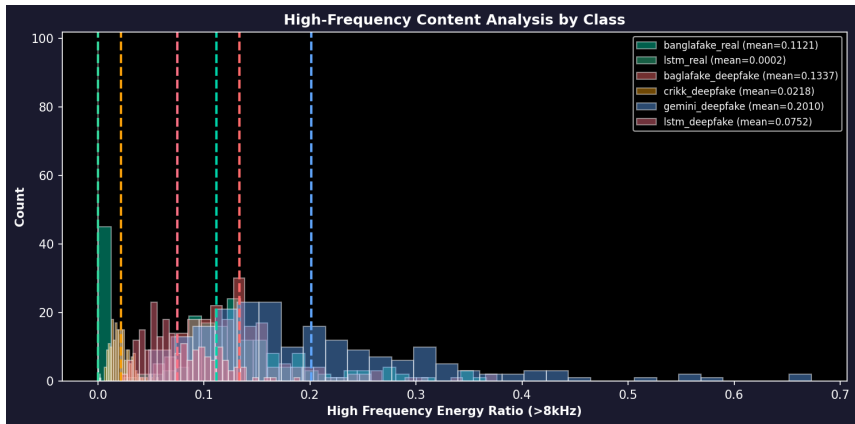


Figure: Augmented Dataset — High Frequency Analysis

t-SNE MFCC Visualization

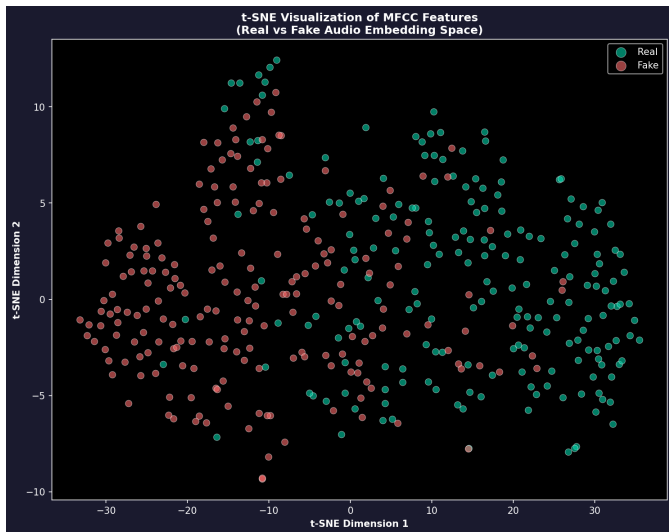


Figure: BanglaFake — t-SNE MFCC

t-SNE MFCC Visualization

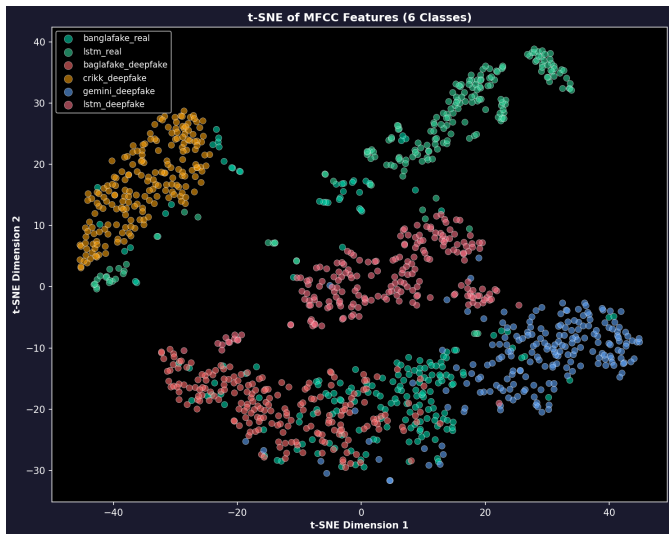


Figure: Augmented Dataset — t-SNE MFCC

Spectrogram Comparison

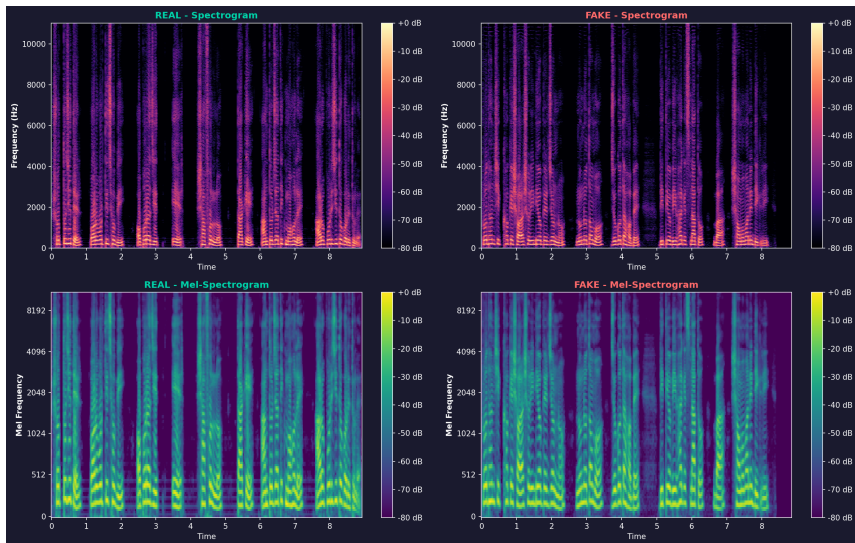


Figure: BanglaFake — Spectrogram

Spectrogram Comparison

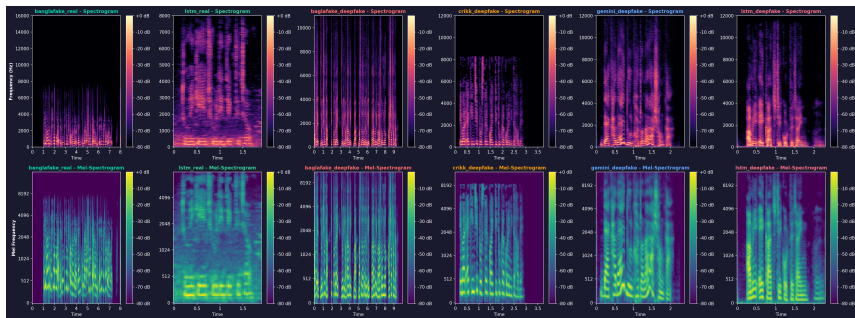


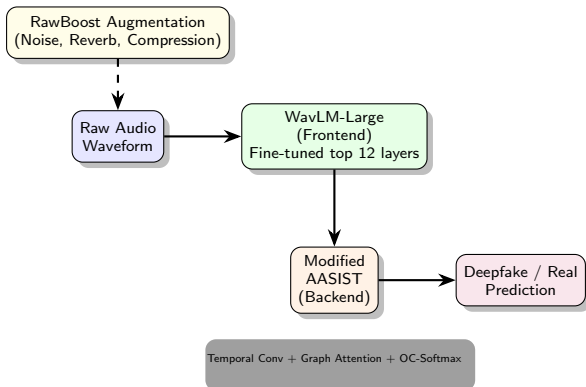
Figure: Augmented Dataset — Spectrogram

Evolution of Models

- **Baseline: MFCC + LSTM**
 - Human-engineered features (loses phase and high-frequency information)
 - Poor generalization to unseen TTS systems
- **Advanced: WavLM-Large + Modified AASIST**
 - Raw waveform input (minimal information loss)
 - Self-supervised frontend + graph attention backend

WavLM + AASIST Architecture

WavLM-Large + Modified AASIST Pipeline



Loss Function: OC-Softmax

Intuition:

- Compacts the "real" class while pushing fake samples away
- Angular margin improves separation in embedding space
- Better suited for detection than standard softmax

$$\mathcal{L}_{OC} = -\log \frac{e^{s(\cos(\theta_y)-m)}}{e^{s(\cos(\theta_y)-m)} + \sum_{j \neq y} e^{s \cos(\theta_j)}} \quad (1)$$

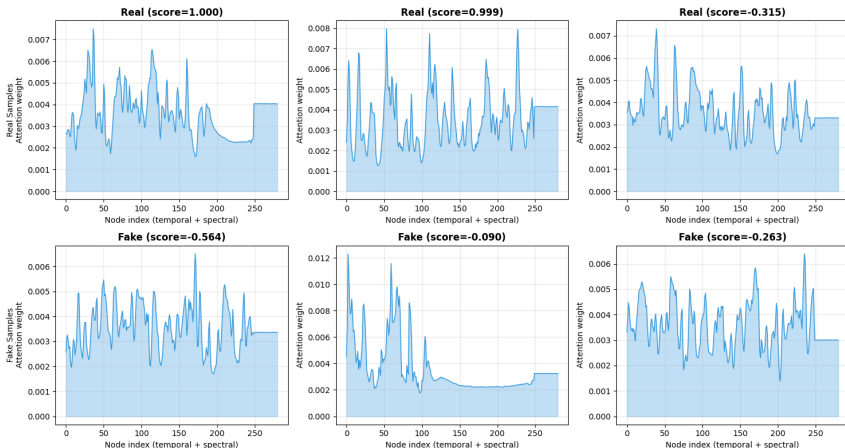
Results on BanglaFake Dataset

Model	Accuracy	EER	ROC-AUC	Notes
LSTM (MFCC)	0.9869	N/A	N/A	Strong on seen VITS
WaveNet (Raw)	0.5000	N/A	N/A	Failed
WavLM + AASIST	0.8062	0.1594	0.9364	Best robustness

Robustness EER:

- 10dB Noise: 0.1837
- MP3 Compression: 0.2223
- Reverb (0.5s): 0.1806

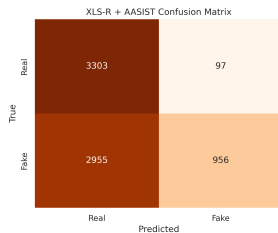
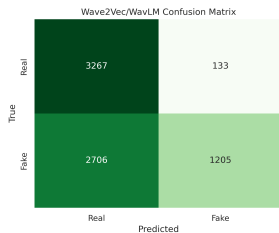
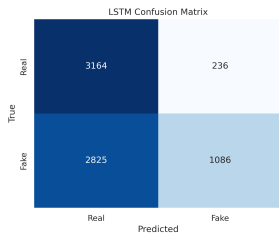
Attention Visualization



Generalization on Augmented Dataset

Model	Accuracy	ROC-AUC	Prec (R/F)	Rec (R/F)	F1 (R/F)
LSTM (MFCC)	0.5813	0.6641	0.5283/0.8215	0.9306/0.2777	0.6740/0.4151
WavLM + AASIST	0.6117	0.8193	0.5470/0.9006	0.9609/0.3081	0.6971/0.4591
XLSR + AASIST	0.5825	0.8328	0.5278/0.9079	0.9715/0.2444	0.6840/0.3852

Confusion Matrix



Ablation Study: Full Dataset Performance

LSTM Model

Scenario	Accuracy	AUC	Precision	Recall	F1-Score
Full Dataset (Stratified)	0.9295	0.9765	0.9312	0.9295	0.9293

WavLM-AASIST Model

Scenario	Accuracy	EER	AUC	Fake Prec	Fake Recall
Full Dataset	0.8522	0.0337	0.9905	1.0000	0.7235

Ablation Study: Holdout Results - LSTM

Holdout Source	Accuracy	AUC	Precision	Recall	F1-Score
lstm_deepfake	0.6592	0.9032	0.7354	0.6592	0.6400
gemini_deepfake	0.6236	0.6842	0.7185	0.6236	0.5929
crikk_deepfake	0.5003	0.7179	0.5766	0.5003	0.4141
Banglafake_deepfake	0.4580	0.3362	0.2144	0.4580	0.2921

Ablation Study: Holdout Results - WavLM-AASIST

Holdout Source	Accuracy	EER	AUC	Fake Prec	Fake Recall
crikk_deepfake	0.5013	0.0679	0.9719	0.9857	0.0690
lstm_deepfake	0.5324	0.0899	0.9512	1.0000	0.1260
gemini_deepfake	0.6265	0.1779	0.9091	0.9964	0.3030
Banglafake_deepfake	0.4644	0.5960	0.3669	0.3333	0.0010

Ablation Study: Summary

Average Holdout Performance

Model	Accuracy	AUC	Notes
LSTM	0.5603	0.6604	Poor generalization
WavLM-AASIST	0.5312	0.7998	21.11% relative AUC improvement

Comparative Analysis

- LSTM achieves higher accuracy on known sources but fails to generalize
- WavLM-AASIST offers superior AUC, very low EER, and near-perfect fake precision
- High precision on fake class makes WavLM more reliable for real-world use
- Trade-off: Slightly lower recall (some fakes may be missed)

Conclusion

WavLM-AASIST is the superior architecture for real-world Bengali audio deepfake detection due to:

- Significantly better generalization to unseen TTS generators
- Higher AUC (0.9905 on full dataset) and much lower EER (0.0337)
- Near-perfect precision when predicting "fake"
- Robust learning of general deepfake artifacts from raw audio

Final Recommendation: WavLM-Large + Modified AASIST with OC-Softmax loss provides a strong foundation for robust Bengali audio forensics.

Current Challenges: The Generalization Gap

- **The BanglaFake Anomaly:** Both LSTM and WavLM-AASIST failed significantly on the BanglaFake holdout ($AUC \approx 0.33 - 0.36$). This suggests a "Generator Drift" where older synthesis artifacts are fundamentally different from modern neural TTS (Gemini/Crikk).
- **The Precision-Recall Dilemma:** While our SOTA model achieved **near-perfect Precision (1.0)** on many holdouts, it suffered from lower Recall (72.35%). In a forensic context, the model is highly "cautious"—it rarely gives false alarms but misses subtle fakes.
- **Environmental Fragility:** Despite using RawBoost, performance EER increased by up to 6% under MP3 compression and Reverb, highlighting the difficulty of "in-the-wild" detection on social media platforms.

Future Directions: Toward Robust Forensics

- **Cross-Generator Ensembles:** Developing a modular architecture where sub-detectors are specialized for different generation families (e.g., Diffusion-based vs. GAN-based synthesis).
- **Domain-Adversarial Training:** Implementing Gradient Reversal Layers (GRL) to force the model to learn features that are invariant to the specific generator, potentially solving the BanglaFake anomaly.
- **Linguistic-Agnostic Pre-training:** Exploring if transfer learning from high-resource languages (English/Mandarin deepfake sets) can enhance the detection of phoneme-level artifacts in Bengali.
- **Real-time Deployment:** Optimizing the WavLM-Large frontend for mobile or edge devices to allow for instant verification of voice messages.

Thank You!

Questions?

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